

Homework Analysis 1

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Problem Statement:

Determine the greatest lower bound (infimum) of the set:

$$A = \left\{ \frac{2x+1}{3y-x} : x, y \in \mathbb{R}, 1 \leq 2x < y \right\}$$

Proof:

We aim to find the infimum i of the set A .

First, we attempt to find a potential candidate for the greatest lower bound of A . To make the fraction $\frac{2x+1}{3y-x}$ as small as possible, we should minimize the numerator and maximize the denominator.

Since $1 \leq 2x$, the smallest possible value of x is $x = \frac{1}{2}$.

Also, since $y > 2x$, we can choose y to be arbitrarily large.

Let $x = \frac{1}{2}$ and consider y approaching infinity. Then:

$$\frac{2x+1}{3y-x} = \frac{2\left(\frac{1}{2}\right)+1}{3y-\frac{1}{2}} = \frac{2}{3y-\frac{1}{2}}$$

As $y \rightarrow \infty$, the denominator $3y - \frac{1}{2} \rightarrow \infty$, so the fraction $\frac{2}{3y-\frac{1}{2}} \rightarrow 0$.

Therefore, the values of A can get arbitrarily close to 0 from above. Hence, we propose $i = 0$ as the infimum of A . Let us now show that $i = 0$ is indeed the infimum of A by Definition 1.3.2. First, we show that 0 is a lower bound for A .

For all $x, y \in \mathbb{R}$ satisfying $1 \leq 2x < y$:

- Since $2x \geq 1$, we have $x \geq \frac{1}{2}$.
- Then $2x + 1 \geq 2\left(\frac{1}{2}\right) + 1 = 2$, so the numerator is at least 2.
- Since $y > 2x \geq 1$, $y > 1$, making the denominator $3y - x > 3(2x) - x = 5x \geq 5\left(\frac{1}{2}\right) = \frac{5}{2}$ positive.

Thus, the fraction $\frac{2x+1}{3y-x}$ is always positive. Therefore, 0 is a lower bound of A .

Now, we show that 0 is the greatest lower bound.

Assume that b is any lower bound of A such that $b > 0$. We will show that this leads to a contradiction, thereby proving that 0 is the greatest lower bound. By the Density Theorem 1.4.3 and its Corollary 1.4.4., given any two real numbers i and b with $i < b$, there exists a real number ϵ such that $i < \epsilon < b$. In our case, since $i = 0$ and $b > 0$, there exists ϵ satisfying:

$$0 < \epsilon < b$$

Our goal is to find an element $a \in A$ such that $a < \epsilon$, which would contradict the assumption that b is a lower bound of A .

Let $x = \frac{1}{2}$ (ensuring $2x = 1 \geq 1$).

We need to find y such that $y > 2x$ and:

$$\frac{2x+1}{3y-x} < \epsilon$$

Compute:

$$\frac{2x+1}{3y-x} = \frac{2}{3y-\frac{1}{2}}$$

We need:

$$\frac{2}{3y-\frac{1}{2}} < \epsilon \implies 3y - \frac{1}{2} > \frac{2}{\epsilon}$$

Solving for y :

$$3y > \frac{2}{\epsilon} + \frac{1}{2} \implies y > \frac{1}{3} \left(\frac{2}{\epsilon} + \frac{1}{2} \right)$$

Since $\epsilon > 0$, the right-hand side $\frac{1}{3} \left(\frac{2}{\epsilon} + \frac{1}{2} \right)$ is a positive real number.

According to the Archimedean Property (Theorem 1.4.2.) we can choose any $y = n \in \mathbb{N} \subseteq \mathbb{R}$ satisfying this inequality and $y > 2x = 1$.

For this choice of x and y , we have:

$$a = \frac{2x+1}{3y-x} = \frac{2}{3y-\frac{1}{2}} < \epsilon$$

since

$$3y - \frac{1}{2} > \frac{2}{\epsilon} \implies \frac{2}{3y-\frac{1}{2}} < \epsilon$$

Therefore, $a < \epsilon < b$, so $a \in A$ and $a < b$, which contradicts the assumption that b is a lower bound of A .

This contradiction shows that no lower bound $b > 0$ exists for A . Hence, 0 is the greatest lower bound (infimum) of A :

$$\inf A = 0$$